

SECTION 2.5

MATRIX FACTORIZATIONS

ELEMENTARY LINEAR ALGEBRA

(MATH 2250-03)

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A factorization of a matrix A is an equation that expresses A as a product of two or more matrices. Whereas matrix multiplication involves combining the effects of two or more linear transformations into a single matrix, matrix factorization amounts to a reversal of this process. In the language of computer science, the expression of A as a product amounts to a *preprocessing* of the data in A , organizing that data into two or more parts whose structures are more useful in some way, perhaps more accessible for computation.

THE LU FACTORIZATION

The LU factorization, described below, is motivated by the fairly common industrial and business problem of solving a sequence of equations, all with the same coefficient matrix:

$$Ax = b_1, \quad Ax = b_2, \quad \dots \quad Ax = b_p$$

When A is invertible, one could compute A^{-1} and then compute $A^{-1}b_1$, $A^{-1}b_2$, and so on. However, it actually turns out to be *more efficient* to solve the first equation in the sequence by row reduction and obtain an *LU factorization* of A at the same time. An LU factorization is a decomposition

$$A = LU$$

for some lower-triangular matrix L and some upper triangular matrix U . While general forms of something like an LU factorizations exist for any matrix, the specific case here assumes that A is an $m \times n$ matrix that can be row reduced to echelon form, without row interchanges. We won't dwell on this too much, however. We also won't spend any time looking at how LU factorizations are derived (though the text has a very nice section explaining an algorithm to do so, if you're interested). At present, we will introduce the LU factorization only as a way to highlight *why* such factorizations are performed.

The main reason finding an LU decomposition of a matrix A might be useful is due to the fact that we can *decompose* the equation $Ax = b$ into the *pair* of equations $Ly = b$ and $Ux = y$. That is, if $A = LU$, then solving $Ly = b$ and $Ux = y$ can be achieved by straightforward back substitution (since L and U are upper and lower triangular), and the equation $Ax = b$ can then be solved by

$$\begin{aligned} Ax &= LUx \\ &= L(Ux) \\ &= Ly \\ &= b. \end{aligned}$$

That is, by breaking A into two parts which are easier to deal with, we can solve equations involving A (like $Ax = b$) much more easily.

Example 1. The matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix}$ admits the LU factorization $A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. Let's use this to solve $Ax = b$, where $b = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$.

First, we solve $Ly = b$. That is, we solve

$$\begin{aligned} Ly &= b \\ \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} &= \begin{bmatrix} 1 \\ 7 \end{bmatrix} \\ \begin{bmatrix} y_1 \\ y_1 + 2y_2 \end{bmatrix} &= \begin{bmatrix} 1 \\ 7 \end{bmatrix}. \end{aligned}$$

So, by back substituting and doing some straightforward algebra, we get

$$y_1 = 1 \quad \text{and} \quad y_2 = (1/2)(7 - y_1) = (1/2)(7 - 1) = 3.$$

So, now we only need to solve $Ux = y$. That is, we solve

$$\begin{aligned} Ux &= y \\ \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 1 \\ 3 \end{bmatrix} \\ \begin{bmatrix} x_1 + 2x_2 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 1 \\ 3 \end{bmatrix}. \end{aligned}$$

By more back substituting and straightforward algebra, we get

$$x_2 = 3 \quad \text{and} \quad x_1 = 1 - 2x_2 = 1 - 2(3) = -5.$$

And now, we are done! In fact, $x = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$ actually turns out to be a solution to $Ax = b$, since $Ly = b$ and $Ux = y$ and $Ax = LUx = Ly = b$. We may verify this in the following ways:

$$\begin{aligned} Ax &= \begin{bmatrix} 1 & 2 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} -5 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} (1)(-5) + (2)(3) \\ (1)(-5) + (4)(3) \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 7 \end{bmatrix} \\ &= b. \end{aligned} \qquad \begin{aligned} Ax &= \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -5 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 7 \end{bmatrix} \\ &= b. \end{aligned}$$